# Latte: Collaborative Test-Time Adaptation of Vision-Language Models in Federated Learning

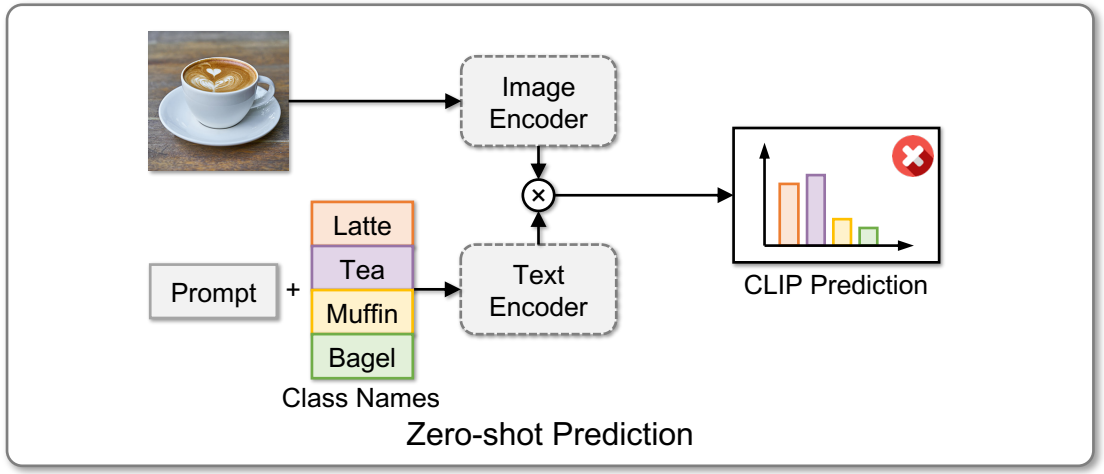
Wenxuan Bao, Ruxi Deng, Ruizhong Qiu, Tianxin Wei, Hanghang Tong, Jingrui He

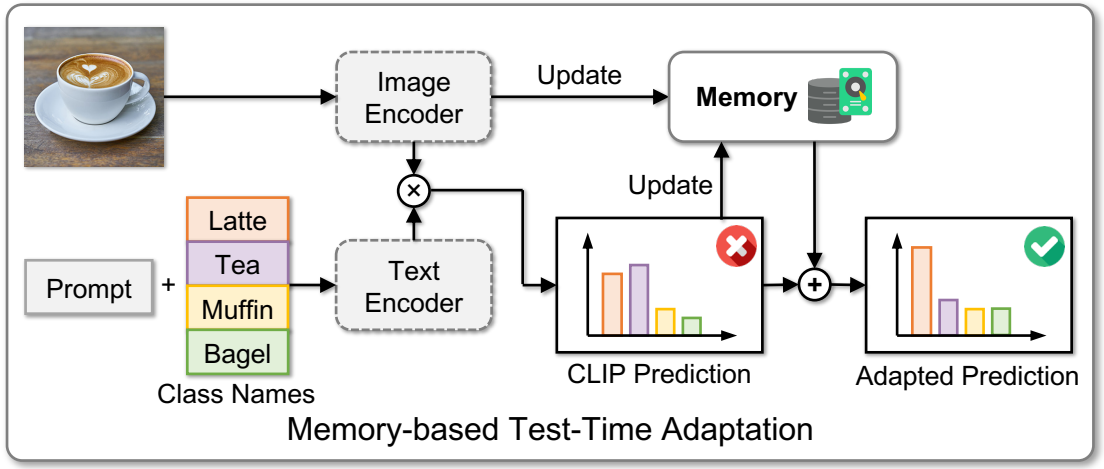
University of Illinois Urbana-Champaign

{wbao4,ruxid2,rq5,twei10,htong,jingrui}@illinois.edu

# Background: Test-Time Adaptation

- **Vision-Language Models** (VLMs) are pre-trained on (image, caption) datasets and possess zero-shot prediction ability.
- **Test-Time Adaptation** (TTA) can improve the performance of VLMs on downstream tasks with domain shifts.
- **Memory-based TTA**. Store high-confidence test image embeddings and pseudo-labels, use them to adjust the classifier based on embedding similarity.
  - For example, saving the prototype for each class $\boldsymbol{C} = [\boldsymbol{c}_1, \cdots, \boldsymbol{c}_c]^\top$, logits= $\boldsymbol{f}^\top \boldsymbol{C}$.
  - Model-agnostic, training-free, good performance with abundant test data.



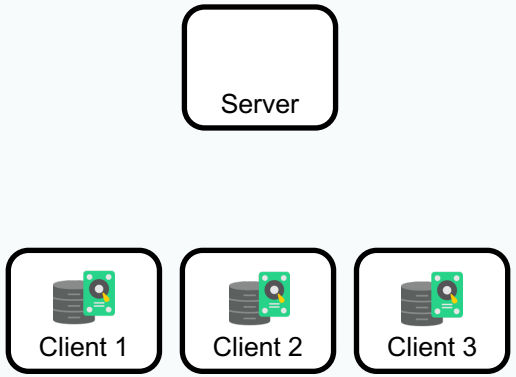
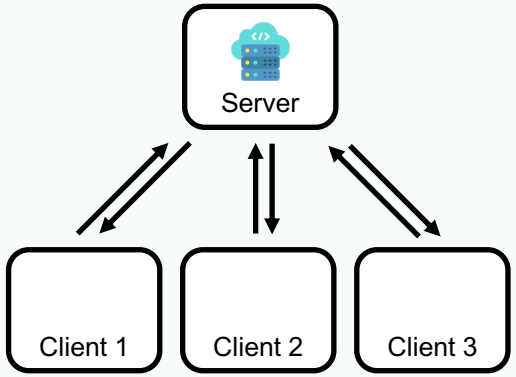
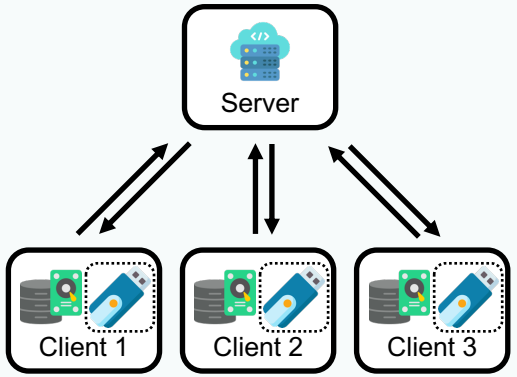





[1] Alec Radford, et al. Learning Transferable Visual Models From Natural Language Supervision. ICML 2021.
[2] Yabin Zhang, et al. Dual Memory Networks: A Versatile Adaptation Approach for Vision-Language Models. CVPR 2024.

# Background: Collaborative TTA

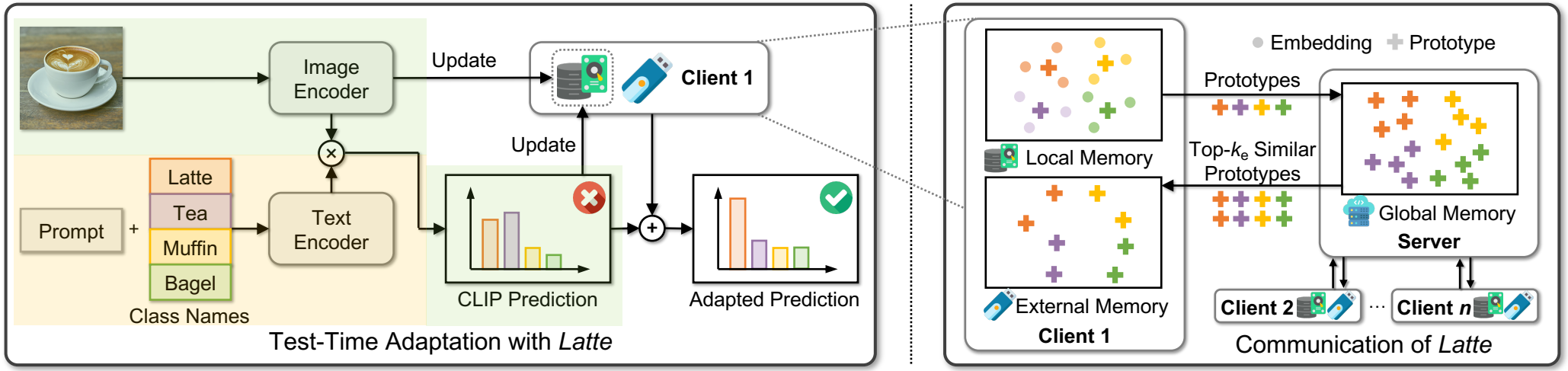
- Federated learning: Adapt to a series of domains with *different* distribution and *limited* data.

| | TTA (local) | TTA (global) | Latte (collaborative TTA) |
|---|---|---|---|
| **Setting** |  <br>Apply TTA to each client independently |  <br>Apply TTA to all clients as a whole |  <br>Apply TTA to each client while sharing information among clients |
| **Memory** | Local memory on each client | Global memory on server | Local + external memories on each client<br>Global memory on server |
| **Personalization** | Client-specific | Not client-specific | Client-specific |
| **Data** | Limited | Abundant | Abundant |

# Proposed Method: Latte

- Each client $i$ has:
  - **Local memory** $\mathbf{L}^i \in \mathbb{R}^{c \times k_l \times d}$ consists of $c$ priority queues $\{\boldsymbol{L}_1^i, \cdots, \boldsymbol{L}_c^i\}$, one for each class.
    - Each $\boldsymbol{L}_y^i = \{\boldsymbol{l}_{y,1}^i, \cdots, \boldsymbol{l}_{y,k_l}^i\} \in \mathbb{R}^{k_l \times d}$ stores up to $k_l$ test image embeddings with pseudo-label $y$, ordered by entropy from low to high. ($d$ is the dimension of embedding.)
  - **External memory** $\mathbf{E}^i \in \mathbb{R}^{c \times k_e \times d}$ consists of $c$ lists $\{\boldsymbol{E}_1^i, \cdots, \boldsymbol{E}_c^i\}$, one for each class.
    - Each $\boldsymbol{E}_y^i = \{\boldsymbol{e}_{y,1}^i, \cdots, \boldsymbol{e}_{y,k_e}^i\} \in \mathbb{R}^{k_l \times d}$ stores the *class prototypes* shared by $k_e$ relevant clients.
- The server has:
  - **Global memory** $\mathbf{G} \in \mathbb{R}^{c \times N \times d}$ stores the *class prototypes* from all $N$ clients.
- *Latte* has four steps:
  1. Encode an input image into image embedding $\boldsymbol{f}$ and get initial prediction.
  2. Update the local memory $\mathbf{L}^i$ with $\boldsymbol{f}$.
  3. Use local memory $\mathbf{L}^i$ and external memory $\mathbf{E}^i$ to get adapted prediction.
  4. (Optional) Communicate with the server and update the external memory $\mathbf{E}^i$.

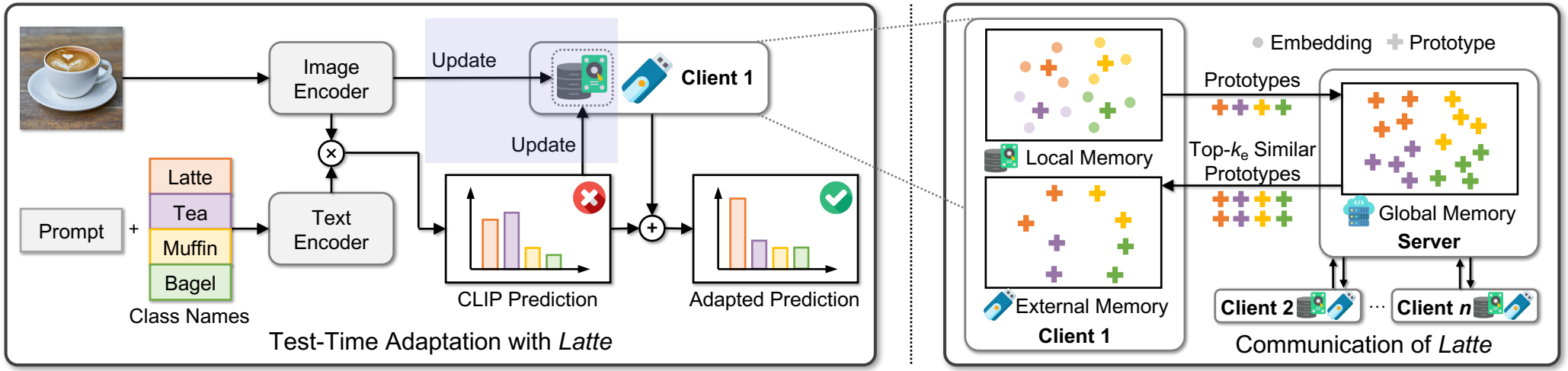
# Latte: Initial Prediction




- **Before testing**:
  - Text embedding $\boldsymbol{T} = [\boldsymbol{t}_1, \cdots, \boldsymbol{t}_c]^\top \in \mathbb{R}^{c \times d}$, where $\boldsymbol{t}_y = \text{text_encoder}(\text{“a photo of \{class } y\text{\}”})$.
- **Initial prediction**:
  - Image embedding $\boldsymbol{f} = \text{image_encoder}(\text{image})$,
  - Initial logits $\boldsymbol{z}_{\text{pre}} = 100 \cdot \boldsymbol{f}^\top \boldsymbol{T}^\top$,
  - Pseudo-label $\hat{y} = \text{argmax}_y \boldsymbol{z}_{\text{pre}}$.

[1] Alec Radford, et al. Learning Transferable Visual Models From Natural Language Supervision. ICML 2021.

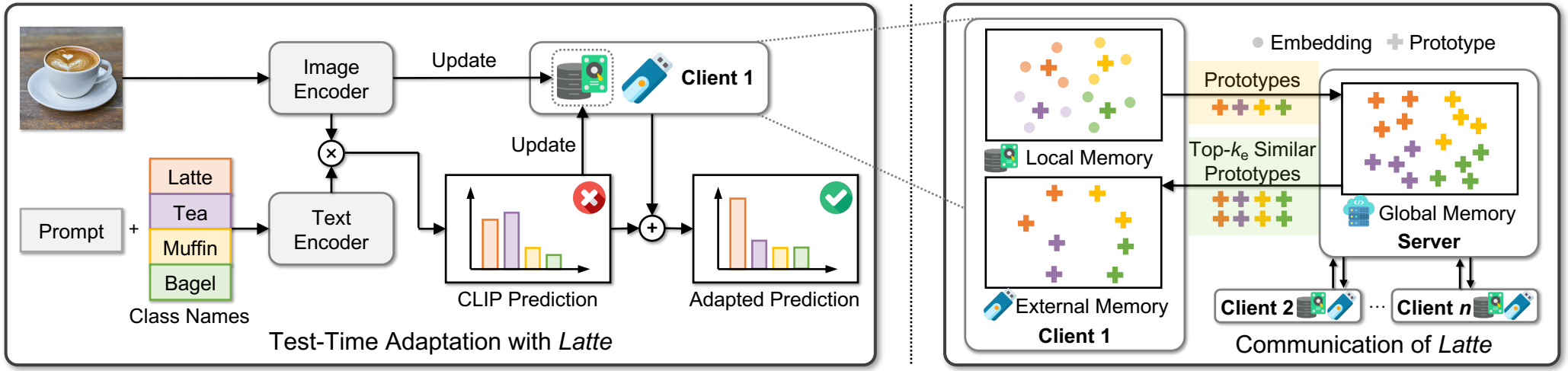
# Latte: Update Local Memory




- **Update local memory**: Maintain the priority queue $\boldsymbol{L}_{\hat{y}}^{i}$ for pseudo-label $\hat{y}$.
  - Compute entropy $H(\boldsymbol{f})$ of new test image's prediction.
  - If $\boldsymbol{L}_{\hat{y}}^{i}$ is not full, the new test image embedding $\boldsymbol{f}$ is directly inserted.
  - If $\boldsymbol{L}_{\hat{y}}^{i}$ is full, compare $H(\boldsymbol{f})$ with max entropy queue; replace it if $H(\boldsymbol{f})$ is smaller.

[1] Yabin Zhang, et al. Dual Memory Networks: A Versatile Adaptation Approach for Vision-Language Models. CVPR 2024.

# Latte: (Optional*) Communication




- **Update global memory**: For each class $y$, compute entropy-weighted class prototype and update client $i$'s corresponding segment in global memory.

$$\boldsymbol{g}_{y,i} \xleftarrow{\text{upload}} \text{normalize}\left(\sum_{\kappa=1}^{k_l} \exp\left(-\gamma \cdot H\left(\boldsymbol{l}_{y,\kappa}^{i}\right)\right) \cdot \boldsymbol{l}_{y,\kappa}^{i}\right)$$

Downweight samples with uncertain prediction.

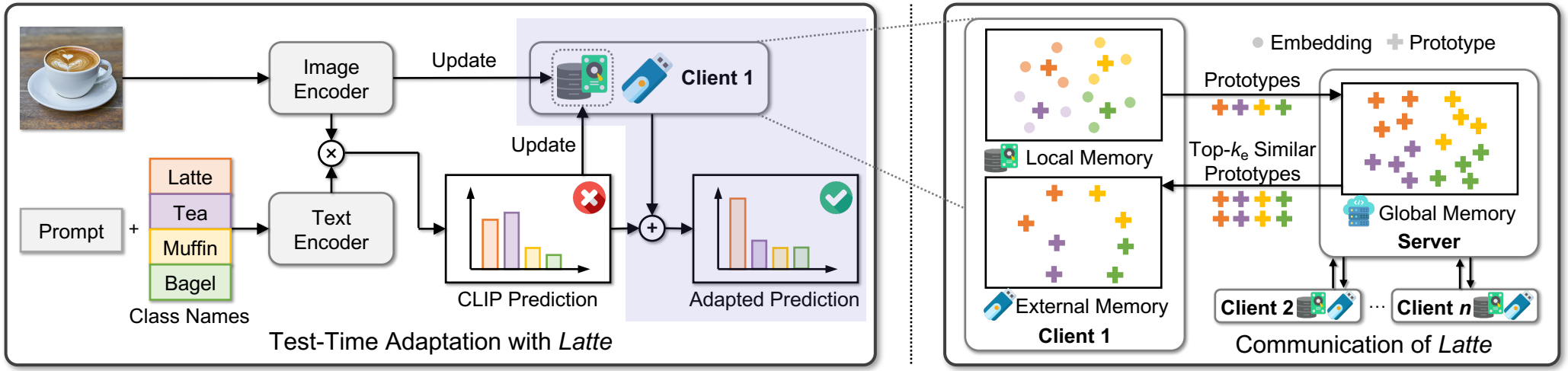
- **Update external memory**: For each class $y$, retrieve top-$k_e$ prototypes most similar to $\boldsymbol{g}_{y,i}$.

$$\boldsymbol{E}_{y}^{i} \xleftarrow{\text{download}} \left\{\boldsymbol{g}_{y,j} \in \boldsymbol{G}_{y} : j \neq i, \boldsymbol{g}_{y,j}^{\top} \boldsymbol{g}_{y,i} \geq \tau(k_e)\right\}$$

Clients with very different distribution will not be retrieved.

*: This step is not required for every inference. It can be executed only when the client connects to the server, supporting asynchronous updates.

# Latte: Adaptation




- Merge local memory $\mathbf{L}^i$ and external memory $\mathbf{E}^i$ into merged memory $\mathbf{M}^i$. For each $y$,

$$\boldsymbol{M}_y^i = \{\boldsymbol{m} \in \boldsymbol{L}_y^i \cup \boldsymbol{E}_y^i : H(\boldsymbol{m}) \leq \tau(k_l)\}.$$

- Use merged memory $\mathbf{M}^i$ to adjust the initial prediction:

Upweight samples more similar to test sample.

$$\boldsymbol{c}_y^i = \text{normalize}\left(\sum_{\kappa=1}^{k_l} \exp(\beta \cdot \boldsymbol{f}^\top \boldsymbol{m}_{y,\kappa}^i) \cdot \exp\left(-\gamma \cdot H(\boldsymbol{m}_{y,\kappa}^i)\right) \cdot \boldsymbol{m}_{y,\kappa}^i\right),$$

$$\boldsymbol{z}_{\text{mem}} = 100 \cdot \boldsymbol{f}^\top \boldsymbol{C}^i, \qquad \text{where } \boldsymbol{C}^i = [\boldsymbol{c}_1^i, \cdots, \boldsymbol{c}_C^i],$$

Downweight samples with uncertain prediction.

$$\hat{y}_{\text{post}} = \text{argmax}_y(\boldsymbol{z}_{\text{pre}} + \alpha \cdot \boldsymbol{z}_{\text{mem}}).$$

# Theoretical Analysis: Generalization

- **Theorem 4.3** (informal). Assume that for each client, there are $n_{\mathrm{ID}}$ *in-distribution (ID) clients* with the same distribution, and $n_{\mathrm{OOD}}$ *out-of-distribution (OOD) clients* with sufficiently different distribution, each providing $N$ test samples. Under distributional assumptions, with probability at least $1-\delta$, the expected error rate on unseen testing samples satisfies

$$\epsilon_{\mathrm{post}} \leq \epsilon_{\mathrm{asym}} + \mathcal{O}\left(\left(\frac{k+\log(1/\delta)}{n_{\mathrm{ID}} \cdot N}\right)^{\frac{1}{d+1}}\right), \text{ where } \epsilon_{\mathrm{asym}} \leq \epsilon_{\mathrm{pre}},$$

where $k$ is the merged memory size, and $\epsilon_{\mathrm{pre}}$ is the error rate of the pre-trained model.

- *Benefit from more data and more ID clients*: Smaller error rate with increasing $n_{\mathrm{ID}}$ and $N$.
- *Robust to OOD clients:* Bound not affected by $n_{\mathrm{OOD}}$.

# Experiments: Effectiveness & Efficiency

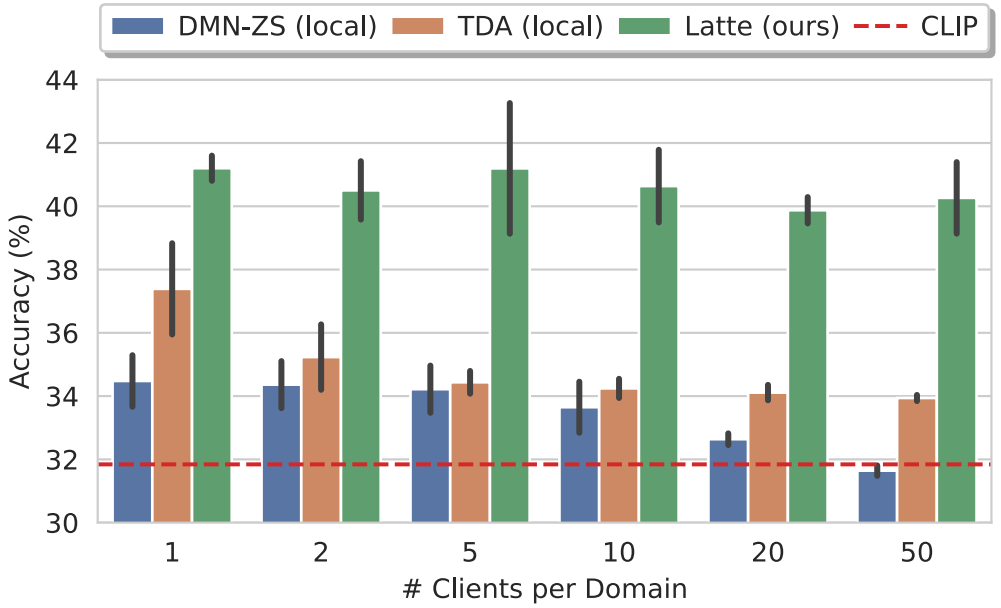
- Latte has good performance on domain adaptation and corruption benchmarks.

| Method | VLCS | TerraIncognita | CIFAR-10-C | CIFAR-100-C |
|---|---|---|---|---|
| CLIP (ViT-B/16) | 80.83 | 31.84 | 65.58 | 38.91 |
| VTE | 81.75 | 38.56 | 67.64 | 37.30 |
| TPT | 81.18 | 35.51 | 65.81 | 37.58 |
| Zero | 81.98 | 38.34 | 67.95 | 37.55 |
| TDA (local) | 81.44 | 34.24 | 66.58 | 39.05 |
| TDA (global) | 80.29 | 36.19 | 65.58 | 38.99 |
| DMN (local) | 81.12 | 33.65 | 67.42 | 39.17 |
| DMN (global) | 80.55 | 37.64 | 63.90 | 36.64 |
| *Matcha* | **82.57** | **40.95** | **68.27** | **39.51** |

- Latte is efficient in computation and communication.

| Method | Computation | | | | | Communication |
|---|---|---|---|---|---|---|
| | Augment | BackProp | MACs | Runtime | Memory | Bytes |
| CLIP | ✗ | ✗ | 17.6G | 8.3ms | 1.5G | (172M) |
| TPT | ✓ | ✓ | 1.72T | 143.2ms | 3.6G | - |
| Zero | ✓ | ✗ | 1.13T | 66.6ms | 2.3G | - |
| DMN-ZS (local) | ✗ | ✗ | 17.6G | 8.5ms | 1.7G | - |
| DMS-ZS (global) | ✗ | ✗ | 17.6G | 8.5ms + RTT | 1.7G | 2.05M |
| *Matcha* | ✗ | ✗ | 17.6G | 8.6ms | 1.7G | 614K |

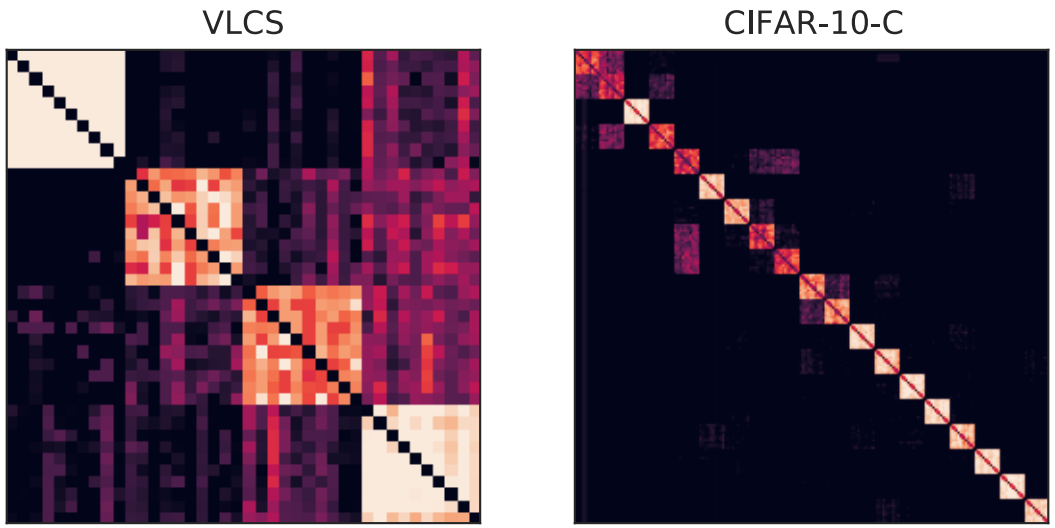
- Local TTA becomes less effective with more clients, as each client has fewer samples for adaptation.
- The performance of Latte is more stable.

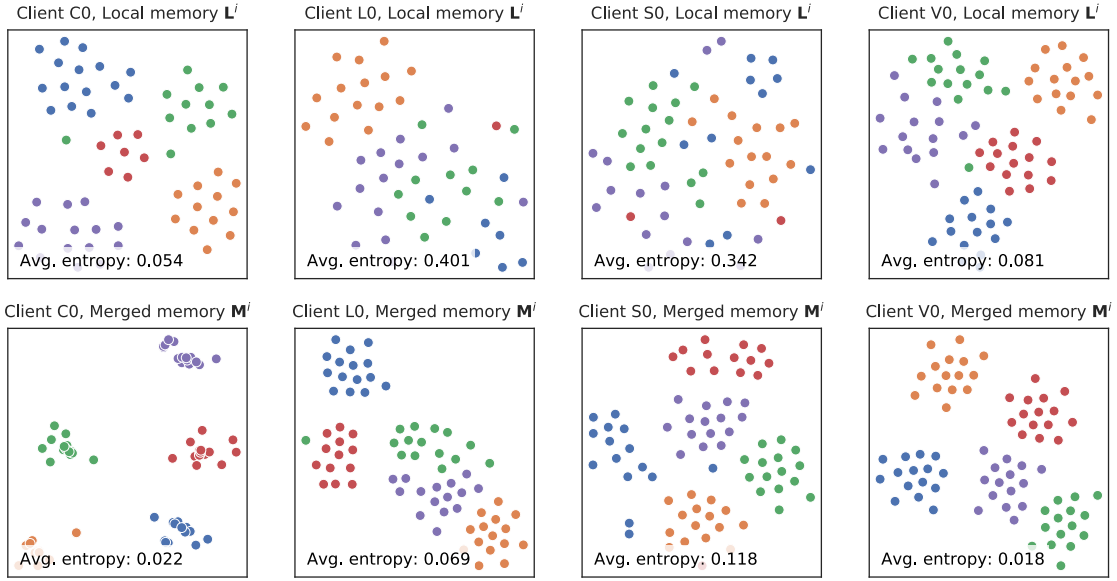


# Experiments: Visualization

Prototype exchanges are more frequent among clients with similar distributions.

The merged memory $\mathbf{M}^i$ has better quality than local memory $\mathbf{L}^i$.



# Key Takeaways

- We propose **Latte**, a collaborative test-time adaptation algorithm for vision-language models in federated learning.
- Latte is a **training-free, memory-based** TTA algorithm.
  - It maintains **local and external memories** on each client, and a **global memory** on the server.
  - The adaptation leverages both **prediction confidence** and **sample similarity**.
- Latte provides a **theoretical guarantee**: post-adaptation performance is no worse than pre-adaptation.
  - Benefits from more samples and more in-distribution (ID) clients.
  - Robust against out-of-distribution (OOD) clients.
- Latte improves accuracy on **domain adaptation** and **corruption benchmarks**, while incurring **negligible additional computation and communication cost**.

# Acknowledgement


- This work is supported by National Science Foundation under Award No. IIS-2416070, IIS-2117902. The views and conclusions are those of the authors and should not be interpreted as representing the official policies of the funding agencies or the government.